Code No: 157DJ **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, January/February - 2023
REAL TIME SYSTEMS
(Common to CSE, IT)

Time: 3 Hours **Max.Marks:75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

1.a)	Differentiate between fork() and vfork()	[2]
b)	Discuss about lseek() command.	[3]
c)	Define a task.	[2]
d)	Discuss various task states.	[3]
e)	What is a signal?	[2]
f)	What are the classification of event types?	[3]
g)	Differentiate between exceptions and interrupts.	[2]
h)	What are spurious interrupts?	[3]
i)	List out the functions of Embedded Linux.	[2]
j)	Discuss about the characteristics of MicroC/OS-II.	[3]

PART – B

(50 Marks)

2.a) Discuss about any six directory Commands in Linux with their functions and examples.
b) Explain i) fork() ii) vfork() iii) exit() iv) wait () v) exec (). [5+5]

OR

3.a) Give an overview of File I/O commands.
b) Give an overview of process control commands. [5+5]

4.a) What is a Scheduler and discuss about its services?
b) Briefly explain different types of scheduling strategies used in RTOS. [5+5]

OR

5.a) Define RTOS? What is the difference between OS and RTOS?
b) Discuss the different operations that can be performed on semaphores. [5+5]

6.a) Explain the I/O subsystem with RTOS examples.
b) Explain the event registers with examples. [5+5]

OR

7.a) Discuss in brief about Pipes and Pipe states.
b) Discuss about i) popen ii) pclose system calls. [5+5]

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8. What is an exception? Explain the process of handling exceptions in RTOS. [10]

OR

9. Discuss about soft timers and various operations of soft timers. [10]

10.a) What is VxWorks?

b) How process management is implemented in VxWorks? [6+4]

OR

11.a) Define porting of RT Linux .

b) Discuss general requirements of processor to port RT Linux along with hardware/software architecture. [5+5]

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